

AI Engineer Take-Home Assignment:

Invoice Processing System

Overview

Build a lightweight AI-powered solution that processes invoices to extract structured information, categorize expenses, and automatically approve or reject them based on predefined business rules.

This is a **prototype assignment**. We value clear thinking and good engineering trade-offs over completeness.

Background

Finance teams receive invoices in various formats (digital PDFs, scanned images, email attachments). Currently, someone manually reviews each invoice, extracts key details, assigns categories, and decides whether to approve or reject based on company policy.

Your task is to automate this workflow while maintaining auditability and transparency in decision-making.

Your Task

Build an end-to-end solution that:

- Ingests invoices (PDF and images support)
- Extracts structured fields from each invoice
- Categorizes invoices into expense categories
- Applies approval guidelines to produce ACCEPT/REJECT decisions with clear explanations
- Outputs results in machine-readable JSON format
- Provides a simple interface (CLI, or minimal web app)

Required Deliverables

A link to [GitHub] repository containing the pipeline code that processes invoices end-to-end.

Technical Requirements

Input Data

Use the invoices provided in the email.

Extracted Fields

Your solution must extract all information you deem essential for the invoice solution.

Approval Guidelines

Implement the following deterministic rules (no "model decides"):

1. **Reject** if total_amount > €500
2. **Reject** if invoice_date < 2017
3. **Reject** if any required field is missing (vendor, invoice_date, buyer's name, total_amount)
4. **Reject** if sum(items.line_total) differs from total_amount by more than €1.00 (tolerance for rounding)
5. **Reject** if individual item_amount > €200
6. **Reject** if the same invoice is submitted more than once
7. **Accept** otherwise

Store approval rules and thresholds in a configuration file. The decision engine must be auditable: each REJECT must include specific reasons tied to which rule(s) failed.

Questions

If you have questions about the assignment:

You can send an email to finanalytics [finanalytics@kpn.com] or Bas Bollaart [basbollaart@kpn.com].

Good luck! We look forward to seeing your solution.